

A5 LDA vs PCA - Answers

LDA vs PCA

1. Which statement correctly distinguishes PCA from LDA?

- ☐ A) PCA is supervised, while LDA is unsupervised
- ☐ B) PCA is unsupervised, while LDA is supervised
- ☐ C) Both are supervised techniques
- ☐ D) Both are unsupervised techniques

Answer: B) PCA is unsupervised; LDA is supervised.

2. What information does PCA primarily use to determine the projection directions?

- ☐ A) Class labels
- ☐ B) Class means
- ☐ C) Variance in the data
- ☐ D) Class probabilities

Answer: C) PCA uses variance in the data.

3. What information does LDA use that PCA does not?

- ☐ A) Feature values
- ☐ B) Class labels
- ☐ C) Number of samples
- ☐ D) Number of features

Answer: B) LDA uses class labels.

4. The primary objective of PCA is to:

- ☐ A) Maximize class separation
- ☐ B) Minimize within -class variance
- ☐ C) Maximize variance in the projected data
- ☐ D) Maximize classification accuracy

Answer: C) PCA maximizes variance in the projected data.

5. The primary objective of LDA is to:

- ☐ A) Maximize total variance
- ☐ B) Maximize between -class separation while minimizing within -class variation
- ☐ C) Minimize the number of samples
- ☐ D) Remove all correlated features

Answer: B) LDA maximizes between-class separation relative to within-class variation.

6. Which technique is generally more suitable when the main goal is classification -oriented dimensionality reduction?

- ☐ A) PCA
- ☐ B) LDA

- o C) K -Means
- o D) Apriori

Answer: B) LDA is designed for label-aware, classification-oriented dimensionality reduction.

7. Which technique can be applied when class labels are unavailable?

- o A) LDA only
- o B) PCA only
- o C) Both PCA and LDA
- o D) Neither PCA nor LDA

Answer: B) PCA can be applied without labels; LDA requires labels.

8. Consider a dataset where the direction of maximum variance does not correspond to the direction that best separates the classes. Which method is more likely to find the class - separating direction?

- o A) PCA
- o B) LDA
- o C) K -Means
- o D) Linear Regression

Answer: B) LDA uses labels to seek a class-separating direction.

9. Which matrices are central to LDA but not required in the same way for standard PCA?

- o A) Within -class and between -class scatter matrices
- o B) Adjacency and distance matrices
- o C) Precision and recall matrices
- o D) Confusion and covariance matrices

Answer: A) LDA uses within-class and between-class scatter matrices.

10. PCA generally finds principal components by performing:

- o A) Eigenvalue decomposition of the covariance matrix
- o B) Classification using class means
- o C) Clustering based on labels
- o D) Regression on target variables

Answer: A) Standard PCA commonly uses eigendecomposition of the covariance matrix (or SVD).

11. LDA commonly obtains discriminant directions using:

- o A) $S_W^{-1} S_B$
- o B) $S_B^{-1} S_W$ only
- o C) The identity matrix
- o D) The confusion matrix

Answer: A) The standard generalized formulation uses $S_W^{-1} S_B$.

12. For a dataset with C classes, the maximum number of meaningful LDA components is:

- o A) C

- o B) $C+1$
- o C) $C-1$
- o D) Equal to the number of features

Answer: C) At most $C - 1$ meaningful LDA components.

13. The maximum number of principal components produced by PCA is generally related to:

- o A) Number of classes
- o B) Number of features
- o C) Number of labels
- o D) Number of class means

Answer: B) PCA components are limited by feature count (and sample rank).

14. Which method attempts to preserve the directions containing the greatest overall variance?

- o A) LDA
- o B) PCA
- o C) Both always
- o D) Neither

Answer: B) PCA seeks directions of greatest overall variance.

15. Which method explicitly attempts to reduce within -class scatter?

- o A) PCA
- o B) LDA
- o C) Both
- o D) Neither

Answer: B) LDA explicitly reduces within-class scatter relative to between-class scatter.

16. Suppose two classes have very different means but their largest variance occurs in a direction where the classes overlap heavily. Which method is more likely to prioritize class separation?

- o A) PCA
- o B) LDA
- o C) Both equally
- o D) Neither

Answer: B) LDA prioritizes class separation using labels, even when it differs from maximum variance.

17. Which statement about PCA and LDA is correct?

- o A) PCA always produces better classification accuracy than LDA
- o B) LDA always produces better classification accuracy than PCA
- o C) Their performance depends on the characteristics and assumptions of the dataset
- o D) PCA and LDA are mathematically identical

Answer: C) Results depend on the data and the methods' assumptions; neither always wins.

18. Which technique is more affected by the absence of class labels?

- o A) PCA

- o B) LDA
- o C) Both equally
- o D) Neither

Answer: B) LDA cannot be fitted as a supervised method without labels; PCA does not need them.

19. A researcher wants to visualize a labeled dataset in 2D while maximizing separation between different classes. Which method would generally be the more appropriate first choice?

- o A) PCA
- o B) LDA
- o C) K -Means
- o D) Apriori

Answer: B) LDA is generally appropriate for labeled class-separation visualization.

20. A researcher wants to reduce the dimensionality of an unlabeled dataset while retaining as much overall variance as possible. Which method should be preferred?

- o A) LDA
- o B) PCA
- o C) Logistic Regression
- o D) QDA

Answer: B) PCA is unsupervised and preserves overall variance.

21. Which statement is NOT correct?

- o A) PCA does not require class labels
- o B) LDA requires class labels
- o C) PCA directly maximizes class separability
- o D) LDA considers class information

Answer: C) PCA does not directly maximize class separability.

22. If the dataset contains 3 classes, what is the maximum number of LDA projection dimensions?

- o A) 1
- o B) 2
- o C) 3
- o D) Equal to the number of original features

Answer: B) Three classes permit at most two LDA projection dimensions.

23. Which method is generally more appropriate as a preprocessing technique when the goal is noise reduction without using class labels?

- o A) LDA
- o B) PCA
- o C) QDA
- o D) Logistic Regression

Answer: B) PCA can reduce noise/variance without class labels, although it does not guarantee noise removal.

24. Which comparison is correct?

- o A) PCA → maximize variance; LDA → maximize class separability
- o B) PCA → maximize class separability; LDA → maximize variance
- o C) PCA → supervised; LDA → unsupervised
- o D) PCA → classification only; LDA → clustering only

Answer: A) PCA maximizes variance; LDA maximizes class separability.

25. If the direction of maximum variance in a dataset is unrelated to class discrimination, what is the main potential disadvantage of using PCA for classification?

- o A) PCA may preserve high -variance features that do not help distinguish classes
- o B) PCA cannot perform dimensionality reduction
- o C) PCA requires class labels
- o D) PCA always removes the class -discriminative information

Answer: A) PCA may retain high-variance directions that do not distinguish the classes.

26. Which technique is more directly designed to find a projection where samples belonging to the same class are close and different classes are far apart?

- o A) PCA
- o B) LDA
- o C) K -Means
- o D) Standardization

Answer: B) LDA explicitly seeks compact same-class projections and separated classes.

27. Which statement best summarizes the difference between PCA and LDA?

- o A) PCA focuses on data variance, whereas LDA focuses on class discrimination
- o B) PCA focuses on class discrimination, whereas LDA focuses on data variance
- o C) Both focus only on reducing computational complexity
- o D) Both require class labels and optimize the same objective

Answer: A) PCA focuses on variance; LDA focuses on class discrimination.

28. In a classification problem, which combination can be useful when the original feature space has a very high number of dimensions?

- o A) PCA followed by LDA
- o B) LDA followed by PCA only
- o C) PCA followed by K -Means only
- o D) PCA followed by data deletion

Answer: A) PCA followed by LDA can first reduce high-dimensionality and then use labels for discrimination.

29. Why might PCA be applied before LDA in a high -dimensional dataset?

- o A) To increase the number of classes
- o B) To reduce dimensionality and help address singularity of the within -class scatter matrix

- o C) To remove class labels
- o D) To convert the problem into clustering

Answer: B) PCA can reduce dimensions and help make S_W nonsingular before LDA.

30. Which pair is correctly matched?

- o A) PCA - Unsupervised - Variance Maximization
- o B) PCA - Supervised - Class Separation
- o C) LDA - Unsupervised - Variance Maximization
- o D) LDA - Unsupervised - Clustering

Answer: A) PCA is unsupervised and maximizes variance.